



SMART INDIA HACKATHON 2026



- **Problem Statement ID - 26134**
- **Problem Statement Title - Challenges in aligning skill development programs with industry requirements and emerging job market demands.**
- **Theme - Miscellaneous**
- **PS Category- Software**
- **Team ID - 121040**
- **Team Name : TEAM S6**





PROPOSED SOLUTION: SKILLX WEBSITE

SkillX is an **AI-powered Labour Market Intelligence & Skill Alignment Platform** that analyzes **real job listings, industry data, and training curriculum** to identify **in-demand, emerging, and rapidly evolving skills**. Using **LLM and machine learning**, it maps **industry skill requirements** against **ITI and MSSDS curriculum** to identify **skill gaps**, measure **curriculum relevance**, and generate **actionable upskilling recommendations**.



Industry-Aligned Course Mapping: Connects **job roles** and **required skills** with relevant **ITI & MSSDS** courses to improve training alignment.



Emerging Skills Identification: Highlights **new and rapidly evolving skills** that are **missing or poorly covered** in **existing curriculum**.



Smart Skill Gap Detection: Uses **Alignment Scores** and **demand insights** to identify the most **critical skill gaps** requiring intervention.



AI-Powered Training Recommendations: Recommends **relevant courses, missing modules, and upskilling pathways** based on industry skill gaps.



INNOVATION AND UNIQUENESS



Job-to-Course Intelligence

Maps **Job Roles, Skills, and Courses** to show how well training prepares candidates for real jobs.



Course Relevance Index

Measures how well **existing courses align with current jobs and industry needs**.



Dynamic Skill Taxonomy

Detects **emerging technologies, evolving skills, and related competencies** to keep the skill database updated.



Demand-Supply Intelligence

Combines **industry demand and training capacity** to identify critical talent shortages.



Demand Velocity

Tracks **current skill demand and its growth or decline** to identify future-ready skills.



Skill Benchmarking

Compares **skill coverage across training programs** to highlight **strengths, weaknesses, and areas requiring improvement**.

01

COURSE AUDIT



Analyzes **ITI & MSSDS** curricula to map existing courses, modules, and skills.



02

MARKET DEMAND ANALYSIS



Analyzes **job listings** to extract **in-demand roles, skills, technologies, and emerging trends**.



03

SKILL MATCH



Maps **industry requirements** against **courses** to calculate **Match Scores** and identify **critical skill gaps**.



04

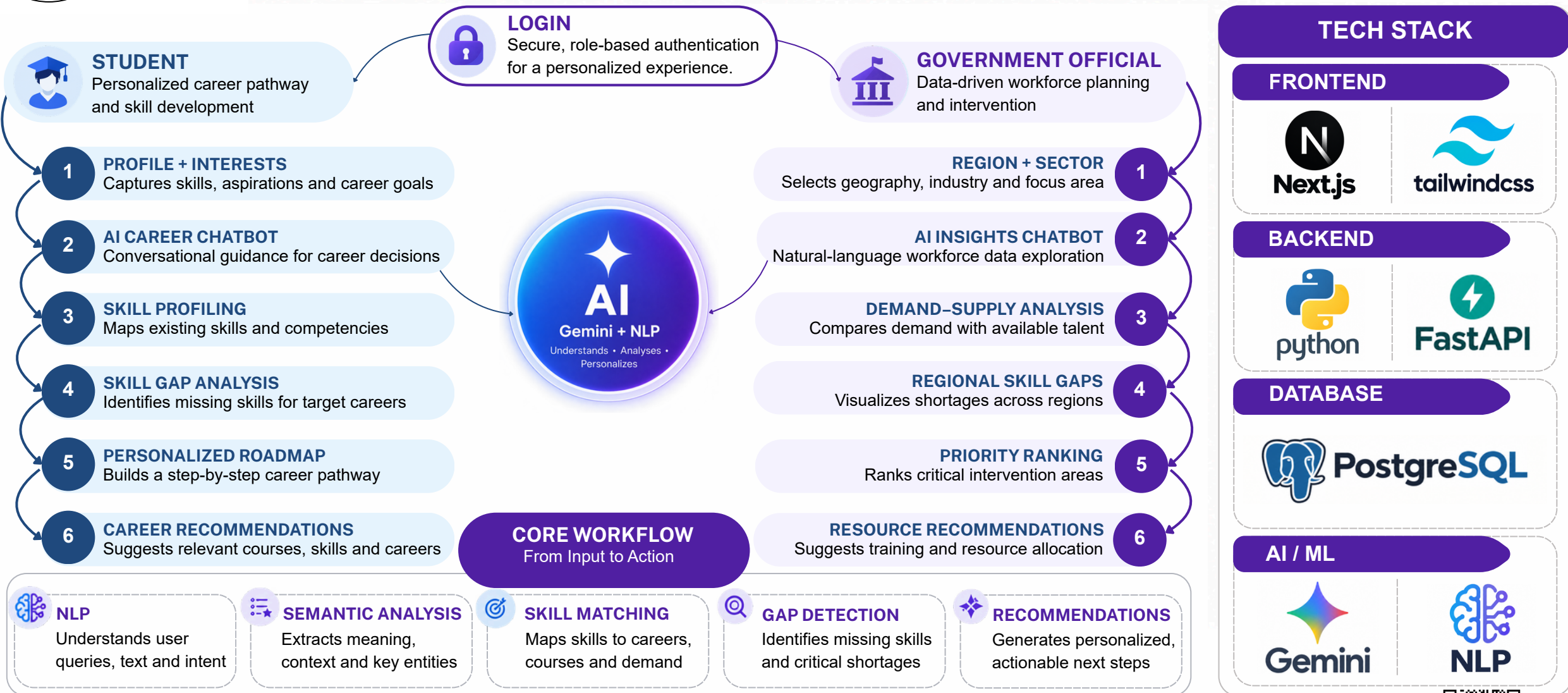
ACTION ENGINE



Converts identified gaps into **curriculum updates, Skill Bridge Packs, and district-wise training recommendations**.



TECHNICAL APPROACH



WATCH THE LIVE PROTOTYPE | SkillIX in Action – Watch the live Demo

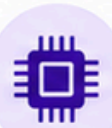
CLICK TO WATCH ↗ OR SCAN



FEASIBILITY AND VIABILITY

FEASIBILITY AND VIABILITY

A practical, evidence-driven approach to continuously align Maharashtra's skilling ecosystem with industry demand.



Proven Technology

Uses established LLMs, ML, APIs, databases and web technologies



Scalable Architecture

Can expand across districts, sectors, courses and states



Labour Intelligence

Combines job, industry and training data to reveal emerging skill demand.



Continuous Updates

Continuously updates recommendations based on changing job demand, skills and industry needs.



Cost Effective

Optimised API usage and minimal infrastructure enable cost-effective scaling.



Growth Ready

Built to cover Maharashtra's districts and evolving workforce needs.



Human Validation

Industry experts validate AI recommendations for practical relevance.



Positive Growth

Improves course relevance, skill alignment, placements and employer satisfaction.



Feasible. Viable. Impactful
Built for Today, Ready for Tomorrow

THE OUTCOME



Industry
Relevant
Training



Better Skilled
Workforce



Higher
Placements



Stronger Industry
Connect



Economic
Growth & Social
Impact

SUSTAINABLE & HIGH-IMPACT



Sustainable Use

Supports continuous curriculum updates and workforce planning as industry needs, job roles and skill requirements evolve.



Shared Benefits

Connects government, institutes, students and employers through shared insights on workforce demand, skill gaps and evolving industry needs for smarter, data-driven workforce decisions.

VIABILITY & SOLUTIONS



Multi-Source Data

Cross-validates diverse data sources to reduce blind spots and strengthen workforce insights.



Validation Layer

Cross-checks AI recommendations with expert and industry inputs for greater accuracy and trust.



Privacy & Security

Protects sensitive data through secure access controls, controlled data handling and responsible AI usage.



Continuous Improvement

Keeps recommendations relevant by continuously learning from new data, feedback and changing workforce needs.

IMPACT AND BENEFITS

SkillX: Creating Impact from Skills to Employment

FOR STUDENTS

- ✓ **Industry-Relevant Skills**
Build skills aligned with current job market demand.
- ✓ **Higher Placement Opportunities**
Target in-demand roles with better-aligned skills.
- ✓ **Personalized Career Roadmap**
Get clear Roadmap based on skills and career goals.
- ✓ **Better Employability**
Job ready skills increase chances of getting hired.
- ✓ **Better Alignment With Emerging Job Roles**
Stay relevant with changing industry trends.
- ✓ **Personalized Course and Skill Recommendations**
AI-Driven suggestions for industrial growth.

FOR EMPLOYERS

- ✓ **Access to Job-Ready Talent**
Skilled workforce ready to contribute from day one.
- ✓ **Reduced Hiring Time and Cost**
Less time spent on screening and upskilling.
- ✓ **Curriculum Aligned With Industry needs**
Training designed with real-time industry input.
- ✓ **Higher Productivity and Efficiency**
Better skilled talent leads to improved performance
- ✓ **Stronger Industry-Acedemia Collaboration**
Continuous feedback loop for mutual growth
- ✓ **Workforce Future-Proofing**
Prepared talent for emerging technologies and roles.

FOR GOVERNMENT & TRAINING INSTITUTES

- ✓ **Evidence-Bases Decision Making**
Data-driven insights for policy and planning.
- ✓ **Timely Course Updates**
Curriculum revised as per latest industry trends.
- ✓ **Optimised Resources**
Better utilisation of funds, equipment and trainers.
- ✓ **District-Level Training Plans**
Localised plans addressing regional skill demands.
- ✓ **Better Placement Outcomes**
Higher success rate through relevant training.
- ✓ **Monitoring & Impact Tracking**
Measure outcomes and improve continously.
- ✓ **Inclusive Skill Development**
Bridge urban-rural skill gaps effectively.

KEY OUTCOMES

-  **Stronger Placement Rates**
Better alignment with training and real job requirement.
-  **Reduced Skill Mismatch**
Identify and address gaps between industry demands and training curricula.
-  **Improved Employer Satisfaction**
Produce candidates with more relevant and job-ready skills.
-  **Faster Curriculum Revision**
Quick updates and removals based on industry feedback.
-  **Data-Driven Skill Ecosystem**
Enables continuous, evidence based planning across districts, sectors, and job roles.



RESEARCH AND REFERENCES

Backed by Credible Sources and Real World Research



GOVERNMENT RESOURCES

1

Ministry of Skill Development & Entrepreneurship (MSDE)

[VIEW SOURCES ↗](#)

2

National Skill Development Corporation (NSDC)

[VIEW SOURCES ↗](#)

3

SWAYAM Plus

[VIEW SOURCES ↗](#)

4

Skill India Digital

[VIEW SOURCES ↗](#)

INDUSTRY & LABOUR MARKET

1

International Labor Organization

[VIEW SOURCES ↗](#)

2

Mercer — Graduate Skill Index 2025

[VIEW SOURCES ↗](#)

RESEARCH

1

57.4% of Indian graduates assessed were not employable

Mercer | Mettl, 2025 [VIEW SOURCES ↗](#)

2

39% of workers' core skills are expected to change by 2030

World Economic Forum, 2025 [VIEW SOURCES ↗](#)

3

63% of employers reports skill gaps as a major barrier

World Economic Forum, 2025 [VIEW SOURCES ↗](#)

OUR DOCUMENTS

1

SKILLX - WORKING PRODUCT VIDEO

[CLICK HERE ↗](#)

2

SKILLX - GITHUB LINK

[CLICK HERE ↗](#)

3

SKILLX - OUR RESEARCHED DOCUMENTS

[CLICK HERE ↗](#)